

SOLAR

SOLAR : A Stepwise Sleep Prediction Framework via Long-Term Anchoring and Refinement



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- **Background**

- Wearable devices and smartphones enable continuous collection of multimodal lifelog data.
- Lifelog data provide rich behavioral and physiological signals for sleep prediction.
- Sleep patterns vary substantially across individuals, highlighting the importance of personalized modeling.
- Sleep outcomes reflect both long-term personal states and day-specific changes.

- **Limitations of Existing Approaches**

- **Short-term focused approaches**
 - Mainly rely on daily observations or short-term statistics.
 - Fail to sufficiently capture long-term behavioral and physiological states.
- **Joint long-/short-term modeling**
 - Encode long-term states and day-specific variations in a single representation.
 - Cannot clearly distinguish their different contributions to sleep outcomes.

Dataset	Description
Activity	Captures the user's daily physical activity and movement patterns.
Screen / App Usage	Represents daily smartphone usage and digital behavior.
GPS	Captures mobility and location-related behavioral patterns.
Wi-Fi / Bluetooth:	Reflects surrounding environments and daily routine patterns.
Light	Measures the user's exposure to ambient light throughout the day.
Charging	Captures smartphone charging behavior and daily usage routines.
Heart Rate	Represents physiological state and its temporal variations.
Step Count	Measures the amount of daily physical activity.

Objective Sleep Indicators

- **S1** : Total Sleep Time
- **S2** : Sleep Efficiency
- **S3** : Sleep Onset Latency
- **S4** : Wake After Sleep Onset

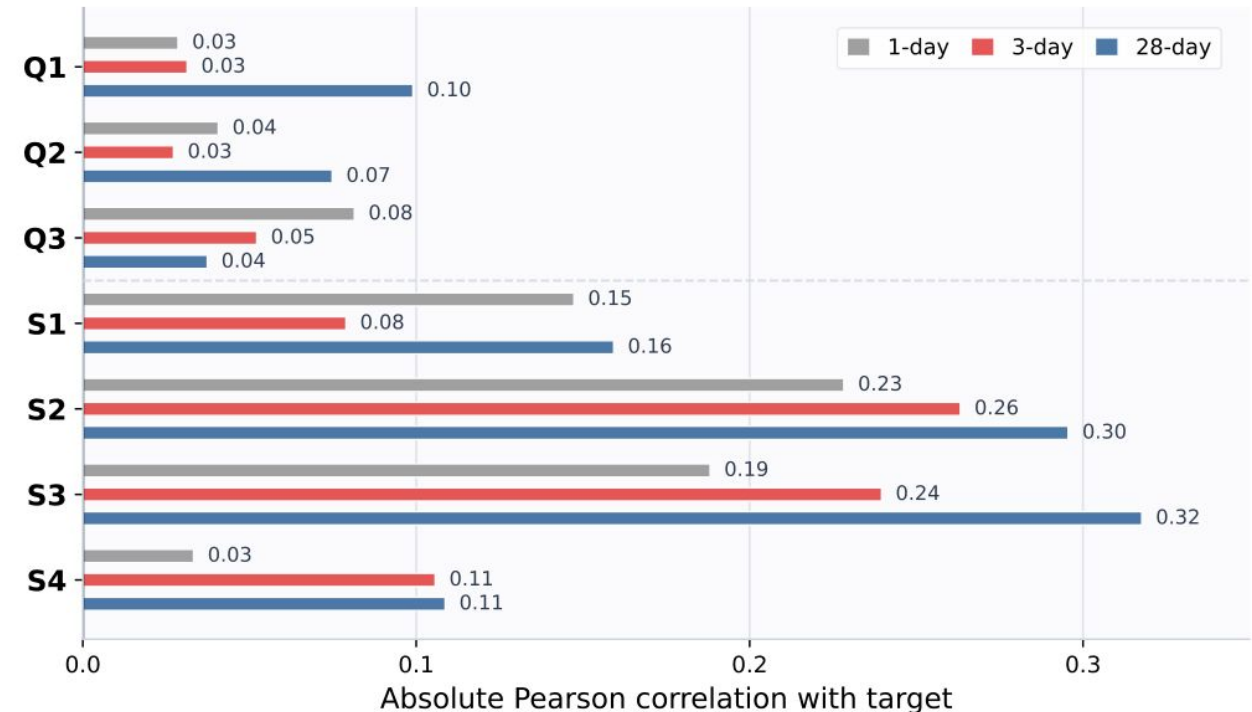
Subjective Sleep States

- **Q1** : Overall Sleep Quality
- **Q2** : Physical Fatigue
- **Q3** : Stress Level

Goal: Predict daily sleep-related outcomes from multimodal personal lifelog data.

- **Observation 1 – Long-term history provides meaningful signals.**
 - 28-day HR patterns show stronger correlations for several sleep targets.
- **Observation 2 – The optimal time scale is target-dependent.**
 - Longer history is not consistently better for every target.
- **Example – Heart Rate (HR)**
 - Compare 1-day, 3-day, and 28-day rolling mean features.
 - Measure absolute Pearson correlation with each sleep target.
 - Example: S3 correlation increases from 0.19 → 0.24 → 0.32 as the history window becomes longer.

Figure 1



- **Long-term Features**

- 14–28 day rolling statistics
- Cumulative mean / standard deviation
- Same-weekday patterns

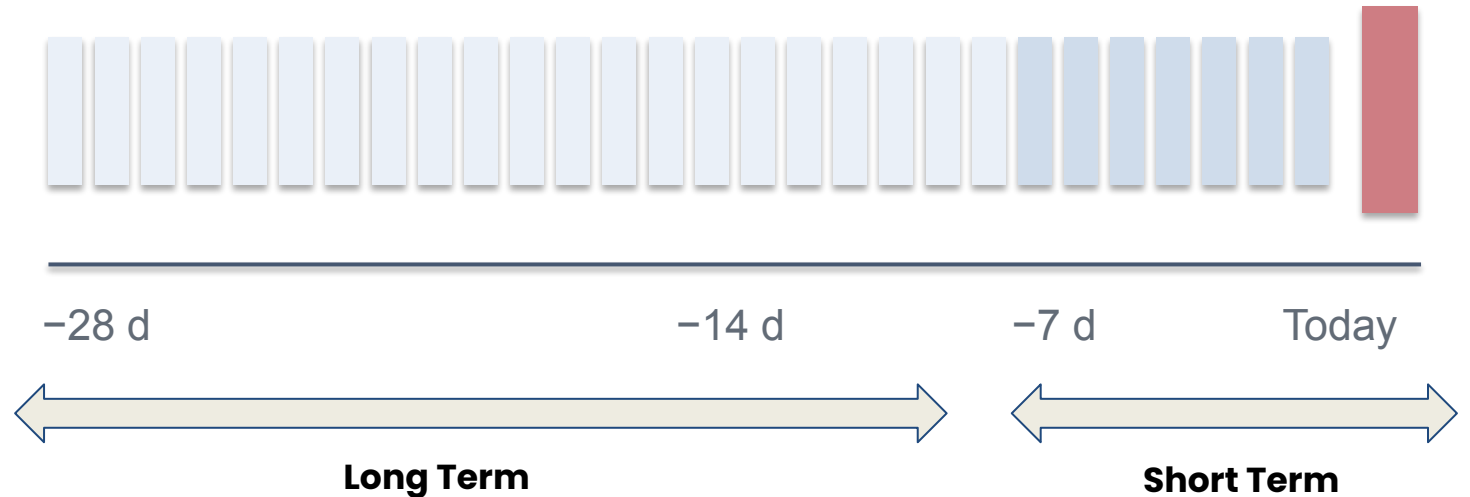
- **Short-term Features**

- 1–7 day recent statistics
- Recent behavioral / physiological changes

- **Personalized Deviation Feature**

- Deviation from long-term baseline Short-long temporal differences
- Individual history & missingness patterns

Accumulated personal history



1. Temporal Anchor

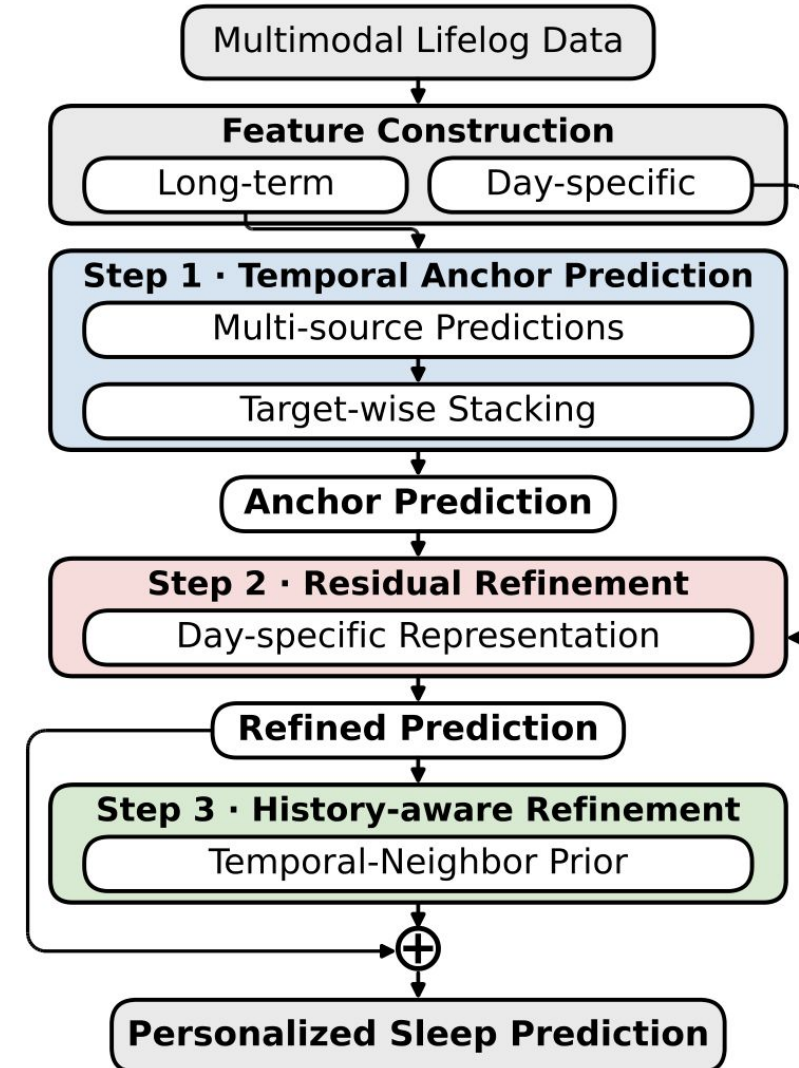
- a. Long-term personal state
- b. Estimate a target-specific personal baseline.

2. Day-specific Refinement

- a. Current state & deviation
- b. Correct transient changes on top of the anchor.

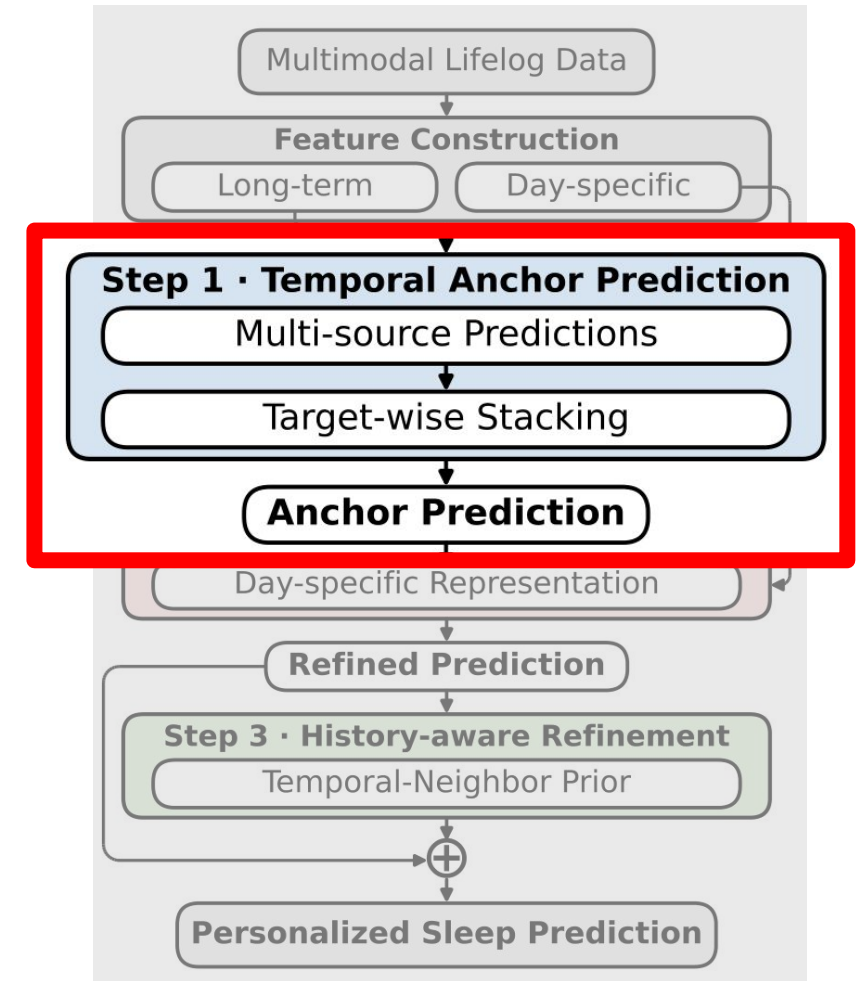
3. History-aware Refinement

- a. Temporal-neighbor history
- b. Refine the prediction only when temporal history helps.



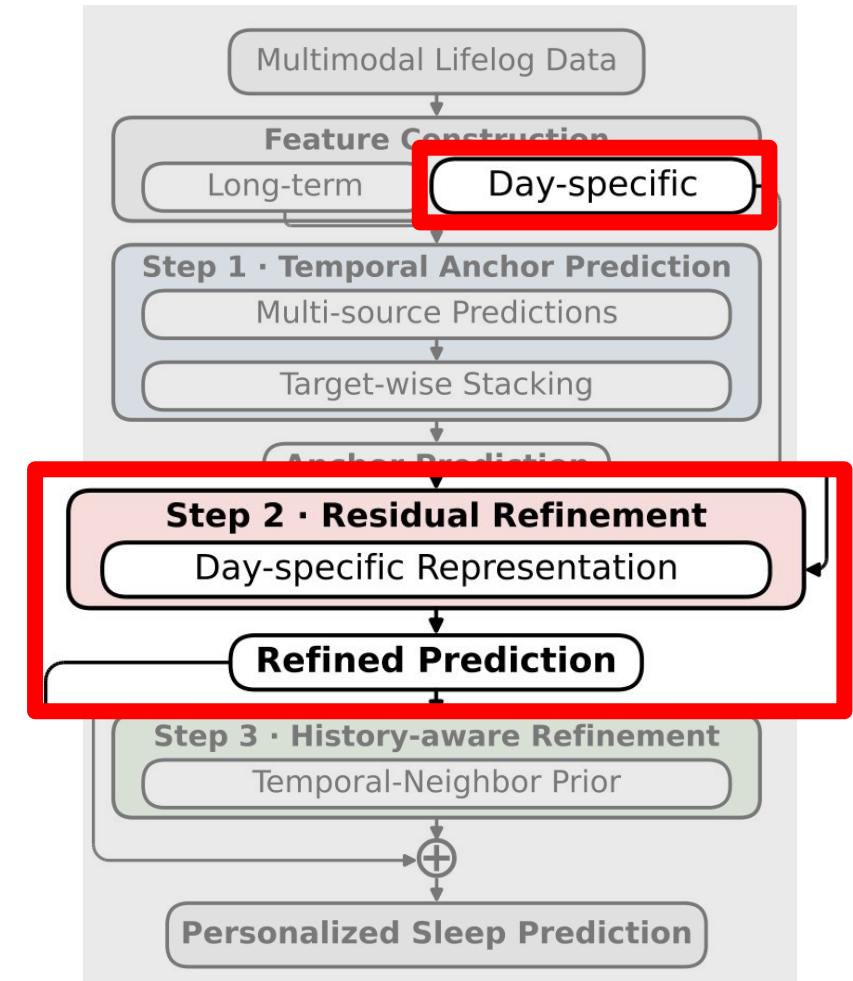
Step 1: Personalized Temporal Anchor

- **GOAL** : Estimate what is typically expected for each participant.
- **Input — Multiple Prediction Sources**
 - Individual history
 - Calendar / weekday patterns
 - Temporal priors
 - Long-term physiological & behavioral features
- **Target-wise Stacking**
 - Different sleep targets rely on different temporal signals.
 - Select useful prediction sources separately for each target.
 - Combine their OOF predictions to construct a robust baseline.
- **Output — Personalized Temporal Anchor**
 - A target-specific baseline probability representing the participant's recurring personal state.



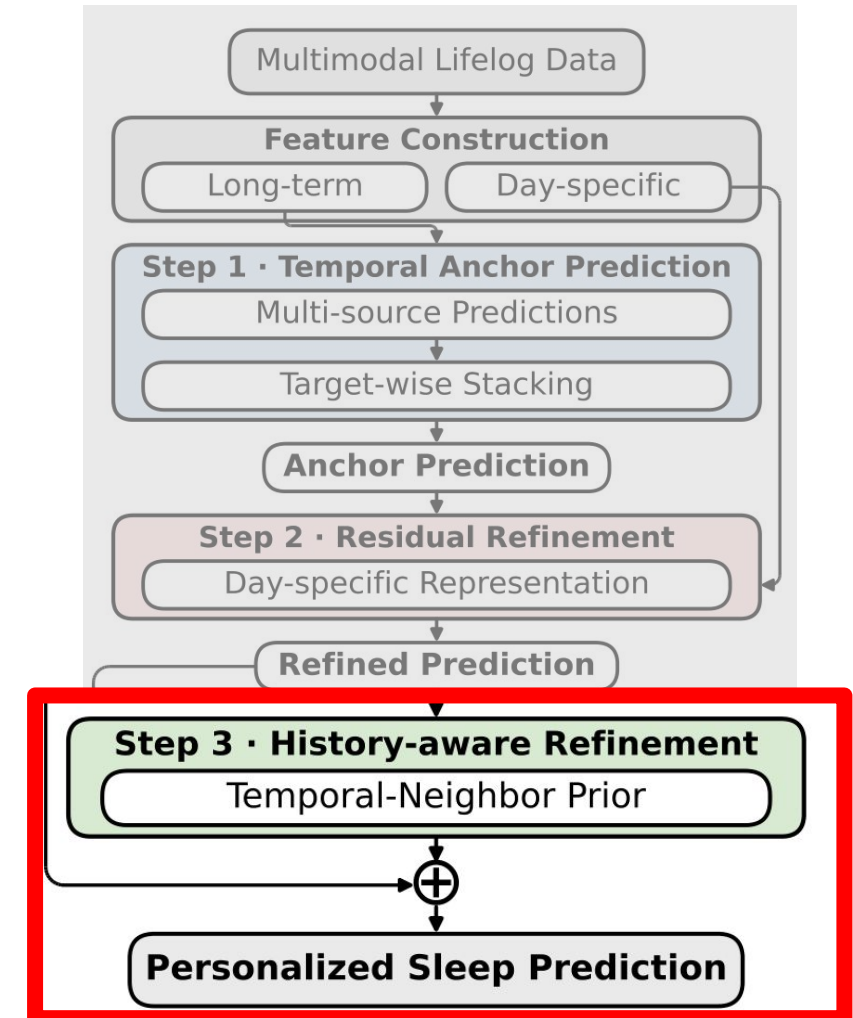
Step 2. Day-Specific Residual Refinement

- **GOAL** : Capture how today differs from the participant's usual state.
- **Day-Specific Information**
 - Current behavioral / physiological state
 - Deviation from the long-term personal baseline
 - Short-term changes observed around the prediction day
- **Residual Refinement**
 - Start from the personalized anchor.
 - Predict only the residual variation not explained by Step 1.
 - Add the correction to the anchor prediction.
- **Controlled Update**
 - Apply a target-specific correction strength.
 - Bound the updated probability through clipping.
- **Output**
 - A refined prediction that preserves personal context while incorporating today's transient changes.



Step 3: History-Aware Refinement

- **GOAL** : Exploit temporal continuity only when it provides additional predictive value.
- **Temporal-Neighbor Prior**
 - Use previous observations from the same participant.
 - Give greater importance to temporally closer observations.
 - Consider both short- and long-term history.
- **Target-wise Validation**
 - Temporal dependency differs across sleep targets.
 - Evaluate whether the history-based prior improves each target.
- **Selective Refinement**
 - If validation improves → apply refinement
 - Otherwise → keep the Step 2 prediction
- **Output**
 - A history-aware personalized prediction without forcing unnecessary temporal correction.



- **Best Overall Performance**
 - SOLAR achieves the lowest average Log-Loss of 0.5560 among all compared methods.
- **Improvement over the Strongest Baseline**
 - Outperforms SCH_CSM (0.6017) by 0.0457, corresponding to a 7.6% relative reduction in Log-Loss.
- **Benefit of the Proposed Framework**
 - The stepwise prediction strategy provides more accurate probabilistic sleep prediction than individual boosting models and their ensemble.

Model	Test (↓)
CatBoost	0.6053
LightGBM	0.6316
XGBoost	0.6985
SCH_CSM (CatBoost:LightGBM = 7:3)	0.6017
SOLAR	0.5560

- **Stepwise Refinement Helps**

- Adding day-specific refinement reduces Test Log-Loss from 0.5642 $\rightarrow$ 0.5626.
- History-aware refinement further improves it to 0.5560.

- **Long-Term Features Matter**

- Removing long-term features increases Test Log-Loss from 0.5560 $\rightarrow$ 0.5739.
- Long-term features provide a 3.1% relative reduction in Test Log-Loss.

- **Consistent Validation Trend**

- The full SOLAR model also achieves the lowest OOF Log-Loss of 0.5760.
- Both ablations support the effectiveness of the proposed design.

Configuration	Validation(OOF) ($\downarrow$)	Test ($\downarrow$)
(a) Stepwise model ablation		
<i>SOLAR</i> (w/o step 2 + 3)	0.5815	0.5642
<i>SOLAR</i> (w/o step 3)	0.5807	0.5626
<i>SOLAR</i>	0.5760	0.5560
(b) Long-term feature ablation		
<i>SOLAR</i> (w/o long-term)	0.5818	0.5739
<i>SOLAR</i>	0.5760	0.5560

- **Short-Term Features Dominate in Total Share**
 - Short-term features account for a larger total SHAP contribution across all seven targets.
 - This partly reflects their much larger feature count.
- **Long-Term Features Are Denser**
 - Long-term features show higher SHAP density for all objective sleep targets (S1–S4).
 - Each long-term feature therefore contributes more strongly on average.
- **Especially Important for Objective Sleep**
 - The density gap is particularly large for S1–S4.
 - Long-term physiological and behavioral states provide informative signals for objective sleep outcomes.

Target	SHAP Share (%)		SHAP Density ($\times 10^{-4}$)	
	Long-term	Short-term	Long-term	Short-term
Q1	1.65	49.89	0.52	0.77
Q2	5.41	68.06	1.79	1.09
Q3	2.20	60.79	0.69	0.92
S1	11.64	37.31	4.14	0.64
S2	16.51	50.18	5.97	0.88
S3	15.99	40.01	5.68	0.69
S4	18.80	53.40	6.13	0.84

Analysis 2: LIME Feature-Level Explanations

- **Target-Specific Feature Patterns**

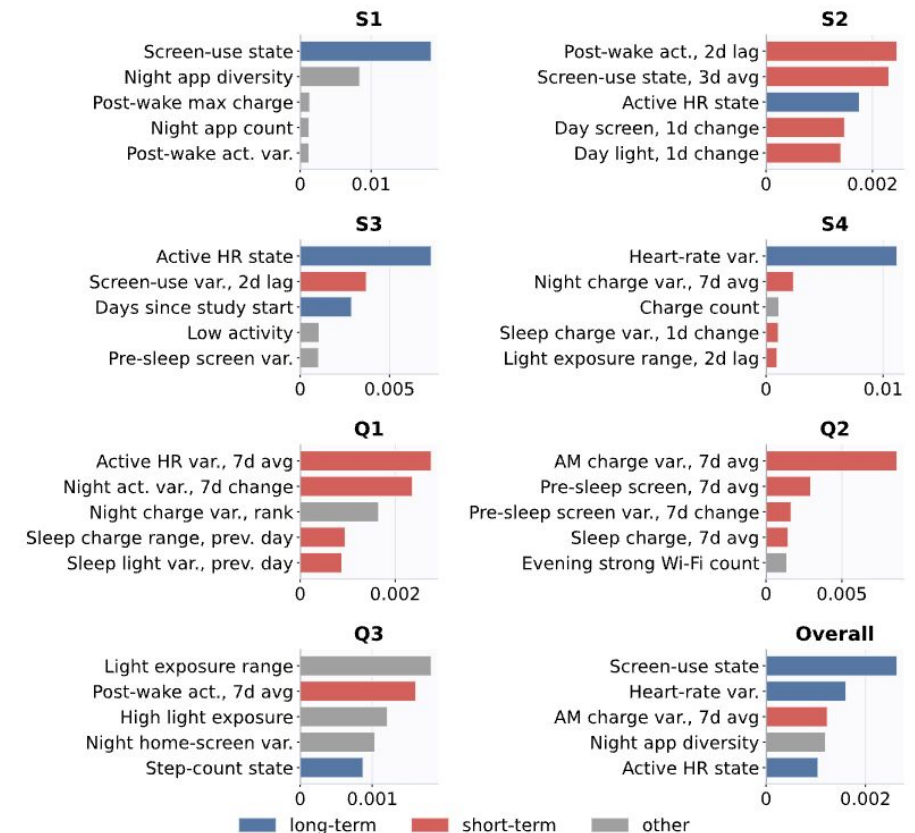
- Important features vary substantially across sleep targets.
- Different targets rely on different behavioral and physiological signals.

- **Long-Term Features Explain Objective Sleep**

- Long-term features are prominent for several objective targets.
- Active HR state is important for S3, while heart-rate variability dominates S4.

- **Long- and Short-Term Signals Are Complementary**

- Subjective and objective targets show different temporal dependencies.
- Both long- and short-term information appear among salient local features.



- **Stepwise Personalized Prediction**

- SOLAR separates long-term personal state, day-specific variation, and temporal-history refinements

- **Performance Improvement**

- SOLAR achieves 0.5560 Average Log-Loss, outperforming the strongest baseline by 7.6%.

- **Long-Term Information Matters**

- Ablation, SHAP, and LIME analyses consistently show that long-term features provide complementary predictive signals.

Thank You

Q & A